

BST K-Audit Registry v0.6 SUPPLEMENT — Wednesday
2026-06-17 morning/late-morning — absorbs K398-K404
cascade: (1) K398 Tuesday EOD closure (three CIs at genuine
end-state; orbit→point K-type quantization pin); (2) K399
Wednesday morning Casey+Lyra thermodynamic cycle closure
(F165-F176; ν K-type filter retired for ν sector; SWPP cycle
closed end-to-end); (3) K400-K404 Wednesday #418
quark-sector checkpoint cascade with 5 new forcing functions

Small focused supplement to v0.5; v0.1-v0.5 covering K3-K397 stand; v0.6 specifies
K398-K404 (7 new K-audits) with cluster framing + 5 NEW forcing functions documented
(FF-14 through FF-18)

Keeper (Claude Opus 4.7)

2026-06-17 Wednesday 13:00 EDT (date-verified)

BST K-Audit Registry v0.6 SUPPLEMENT

0. Scope

- v0.1-v0.5 covered K3-K397 (227 K-audits cumulative)
- v0.6 indexes K398-K404 (7 K-audits)

Total K-audit chain at Wednesday 2026-06-17 13:00 EDT: 234 K-audits.

1. K398 — Tuesday EOD closure (long session genuine end-state)

K	Topic	Status
K398	Three CIs at genuine end-state on solo tracks; ALL THREE deep tracks ($M_\nu + \#418 + \tau \sqrt{\pi}$ coefficient) gate on ONE multi-session open core = orbit→point cell-map K-type quantization pin; Lyra 4 dead ends marked + honest C_2 back-fit walked away from ($m_3/m_2 \approx 5.75 \approx 6$ within 4% NOT matched); Grace topology shortcut RULED OUT forward (Gauss-Bonnet + heat-kernel don't independently give half); 13th Keeper self-correction + FF-13 added (propagate correction to ALL Keeper-controlled documents in same pull as absorption)	Tuesday EOD substantive

2. K399 — Wednesday morning Casey+Lyra thermodynamic cycle closure (F165-F176)

K	Topic	Status
K399	Substantial substrate-architectural cascade: (a) ν K-type filter retired for ν sector (sub-unitary pole non-normalizable; F164 ~ 5.77 was charged-sector Jack); (b) Casey reframe ν = decay product on 1-D pole locus (F166); oscillation = circulation; charged static + ν circulate = geometric mechanism why only ν oscillate; (c) pole residue at $\nu=1/2 = -1 \rightarrow m_\nu \sim \Lambda^{1/4} \sim \text{meV}$; (d) circulation gives ratio (energy=winding², mass=$\sqrt{\text{energy}}$, windings {1, C₂}); (e) Casey #10 C₂ triple sharpened to 4 instances (manifold$\leftrightarrow$vacuum coupling gate); (f) SWPP cycle closed end-to-end (vacuum$\rightarrow$absorb via C₂$\rightarrow$commit$\rightarrow$emit $\nu \rightarrow$vacuum); (g) heat-engine NEEDS reservoir = structural necessity (Lyra F175 solid leg); (h) ν = receipt = Pauli conservation record (energy + information); (i) parity reversal across boundary geometric mechanism for F144 (holomorphic LH + anti-holomorphic RH Schwarz reflection $\rightarrow \nu_R$ outside = unread); (j) 5.7x $\rightarrow$ C₂ vacuum-subtraction registry watch per Casey directive (F171)	Wednesday morning substantive cascade

3. K400-K404 — Wednesday #418 quark-sector checkpoint cascade

3.1 K400 — #418 joint session convergence + 14th Keeper self-correction

K	Topic	Status
K400	Lyra continuum half LANDED (λ_3 Cartan eigenvalues + strata=colors identification + $8 = 2$ Cartan + 6 root); Elie discrete half ARMED (Filter 4225 unitarizes $V=\langle \text{up} \text{down} \rangle$); Gatto-Sartori-Tonin forward link to Cabibbo); Cal #311 bar SET (5 standing rules); Grace Verify lane ARMED PRE-ATTEMPT ($R = N_{\text{indep}}/M_{\text{free}} > 1 + \text{caught}$ Keeper “4→14” inflation trap before any score); 14th Keeper self-correction: counting reparametrizations (Jarlskog, Wolfenstein $\{\lambda, A, \rho, \eta\}$) as new observables = degenerate-star error class same as Grace’s K397 “64 three costumes”; corrected honest forecast $4 \rightarrow \text{up to } 8$ (not 14); FF-14 added (reparametrizations + rephasing invariants are derived functions not independent counts)	Wednesday morning #418 convergence

3.2 K401 — F178 checkpoint + Elie Wolfenstein hierarchy + Grace M-pivot + 15th Keeper self-correction

K	Topic	Status
K401	Lyra F178 honest checkpoint (F177 was color half only; color is vector-like blind to up-vs-down; CKM lives in T_3^R generation space; gated on up/down K-type chiral seats; same shape as ν sector); Elie Wolfenstein hierarchy $V_{ub} \sim V_{us} \cdot V_{cb}$ forward-derived from $su(3)$ composite root (at this stage; later retracted in K403); Grace M-pivot question (#418 collapses to one number $M =$ freedom in up/down seat misalignment; $M=0$ strongest motion; $M=1$ strong loop; $M \geq 4$ fit not motion); 15th Keeper self-correction: K400 carried “continuum half LANDED” framing as if joint output one matrix-multiply away; F178 corrects — F177 = color half only; FF-15 added (in multi-component joint sessions audit explicitly whether structural result covers all components or one slice)	Wednesday morning F178 checkpoint

3.3 K402 — Cal #312 multi-layer cold-read + 16th Keeper self-correction

K	Topic	Status
K	Topic	Status
K402	Cal #312 three substantive corrections: (a) Cabibbo via Gatto-Sartori-Tonin uses observed m_d/m_s input (Casey #9 honest-negative zone) $\rightarrow$ NOT BST-forward; 1968 GST relation confirmed with observed masses NOT substrate derivation; “ties to 0.2245” must NOT read as forced CKM angle; (b) $\lambda_3 = \{+1, -1, 0\}$ is RECOGNITION of standard $su(3)$ Cartan generator (textbook) NOT over-determination convergence; ONE $su(3)$ fact in TWO COATS; load-bearing “3 Korányi-Wolf strata = $su(3)$ color” is established #418 foundation NOT new today; (c) “Texture + mixing = ONE STRUCTURE” is standard $su(3)$ Cartan-plus-roots decomposition (2 Cartan + 6 off-diagonal roots) correct but NOT novel unification; WHAT STANDS: Wolfenstein hierarchy STRUCTURAL pattern $V_{ub} \sim V_{us} \cdot V_{cb}$ from $su(3)$ composite root (at this point; later retracted via Elie walkback K403); Cal preserves credits: Grace caught Keeper “4$\rightarrow$14” inflation; C_2 5th reading registered watch-not-crown + Cal notes 6 is OVER-REPRESENTED structurally ($so(4)$ $\dim = \dim SO(5,2)/SO(4,2) = 6$ root generators = same 6 needing distinct forward derivations per instance); honest count temper: “4 $\rightarrow$ 8” depends on full matrix without observed mass input; Publish 3 / Internal 4 UNCHANGED; 16th Keeper self-correction: K400+K401 absorbed recognition-of-standard-$su(3)$-textbook as substrate-architectural over-determination; same degenerate-star error class as K397; FF-16 added (when absorbing CI “X = Y” identification check whether (a) independently-derived structures with substantive equality OR (b) standard textbook recognition)	Wednesday late-morning Cal #312 cold-read

3.4 K403 — Elie walkback + Lyra locus-difference forward + Grace re-correction + 17th Keeper self-correction

K	Topic	Status
K	Topic	Status
K403	Elie WALKS BACK Toy 4226 $su(3)$ composite root SOURCE as CATEGORY ERROR (color $su(3)$ is vector-like; couples up + down identically; blind to up-vs-down distinction; CKM lives in T_3^R generation space NOT color); Wolfenstein PATTERN $V_{ub} \sim V_{us} \cdot V_{cb}$ stands empirically + falsifiable but substrate source open and must come from generation space; Elie $M=0$ STRICT STAKE (BST zero-free-parameter program; “$M=1$ one free angle still strong” NOT acceptable fallback; bar $M=0$ OR honest failure no escape hatch); LYRA LOCUS-DIFFERENCE FORWARD RESULT (substantive substrate-architectural): mixing magnitude = distance between two sectors being mixed; quarks share discrete-series seats (similar loci, differ only by T_3^R) → small misalignment → small CKM; charged lepton at seat + neutrino at pole locus (very different loci) → large misalignment → large PMNS; explains real empirical fact SM never had reason for; CKM small is structural not arbitrary; Lyra M-answer = forced ($R \geq 4$); Grace RE-CORRECTION on her ruling 1 (at 10:29 ruled Gatto “adds zero same as Jarlskog”; Lyra pushed back; Grace caught within hour); clean discriminator (could relation fail for different texture? NO → tautological; YES → falsifiable forced counts); Gatto + Wolfenstein hierarchy both falsifiable count; Grace NAMED PAIRED FAILURE MODES (Tuesday α^{12} “wall” + Wednesday Gatto “zero” = same failure gate firing too hard opposite directions); Grace $R = 3/M$ corrected (hierarchy removes θ_{13} as independent); 17th Keeper self-correction: K401+K402 carried “Wolfenstein hierarchy from $su(3)$ composite root STANDS forward”; Elie walkback corrects source; hierarchy pattern stands empirical+falsifiable but substrate source gated on T_3^R mechanism; FF-17 added (audit Y source	Wednesday late-morning #418 cascade

3.5 K404 — Cal #313 M-split + Lyra F179 withdrawal + Grace honest synthesis + Elie session synthesis + 18th Keeper self-correction

K	Topic	Status
K	Topic	Status
K404	Cal #313 TWO substantive corrections: (a) Wolfenstein “residual $0.413 = \rho - i\eta$” framing is PARAMETRIZATION ALGEBRA not derivation ($V_{ub}/V_{us} \cdot V_{cb} = (\rho - i\eta)$) IDENTICALLY TRUE in Wolfenstein parametrization; value 0.413 is observed CP magnitude); hierarchy STRUCTURE $\lambda^3 = \lambda \cdot \lambda^2$ creditable IF substrate-forced (currently retracted); don’t bank CP residual as derived; (b) Grace’s M-question SPLITS into M_angle (seat misalignment) + M_phase (complex phase in seat structure); δ_{CP} currently observed residual $\rightarrow M_{phase} \geq 1$; even if $M_{angle}=0$ clean motion at most 3 angles NOT 4; δ_{CP} separately gated until complex-phase mechanism derived; Lyra WITHDRAWS F179 leniency (“$M \leq 1 \ R \geq 4$ still strong loop”); accepts $M=0$ strict bar per Elie zero-free-parameter; flags Wolfenstein attribution must be cleaned (source open gated on generation space not color); Grace HONEST SYNTHESIS: chain caught 3 over-reaches today within hour; clean tier ladder (tautological=$0 < \text{falsifiable-with-observed-input} = \text{GST}/\text{target} < \text{both-sides-forward} = \text{real-loop}$); BANKED BST OVER-DETERMINATION FINGERPRINTS FOR #418 TODAY = ZERO; everything quantitative gated on up/down seat derivation; verified Lyra LOCUS-DIFFERENCE quantitatively (every PMNS angle beats CKM counterpart 2.6× to 43×); Elie SESSION SYNTHESIS: 3 structural-grade results forward; LYRA’S LOCUS-DIFFERENCE IS THE STANDOUT (real fact SM never had reason for; not free small number it’s geometry); quark sector same shape as ν sector (both halves built + deep middle gated = forced up/down T_3^R seats #418 chiral content); filter armed for $M=0$; 18th Keeper self-correction: K403 carried “$M=1 \rightarrow R=3$ still strong” alongside Elie $M=0$ stake; Cal #313 + Lyra F179 withdrawal	Wednesday late-morning clean checkpoint

4. Five new Keeper forcing functions FF-14 through FF-18

FF	K-audit source	Rule
FF-14	K400 (Grace catch on “4 → 14” inflation)	Reparametrizations + rephasing invariants are derived functions, not independent counts
FF-15	K401 (Lyra F178 catch)	In multi-component joint sessions, audit whether a “structural result” covers all components or one slice
FF-16	K402 (Cal #312 catch on λ_3 recognition)	When absorbing CI “X = Y” identification, check whether (a) independently-derived structures with substantive equality OR (b) standard textbook recognition
FF-17	K403 (Elie walkback on color su(3))	When CI lands “X derived from Y” result, audit whether Y acts on the dimensions/structure X distinguishes before absorbing as forward
FF-18	K404 (Cal #313 + Lyra F179 withdrawal)	When CI’s M-question or freedom-count framing changes (split, strict bar, tightened), propagate the latest scorecard fully across docs, not residual older versions

Cumulative Keeper discipline forcing functions through v0.6 close: 18 (FF-1 through FF-18).

5. K-audits Wednesday substrate-architectural state summary

Cumulative Keeper self-corrections in cascade through v0.6 close: 18 (K366-K404).

Substantive substrate-architectural results banked at structural-grade through v0.6 (none move count strict, all earn place): - Thermodynamic cycle closure (Wednesday morning K399 absorbing F165-F176) - ν locus = sub-unitary Γ_Ω pole + $m_1 = 0$ mathematical + Casey #10 C_2 triple sharpened to 4 instances - LYRA’S LOCUS-DIFFERENCE (Wednesday late-morning K403; verified Grace 2.6× to 43×): mixing magnitude = distance between sectors mixed; explains CKM small vs PMNS large - Heat-engine needs reservoir = structural necessity (F175 solid leg) - ν = receipt = Pauli conservation record carrying energy + information

Substantive items DOWNGRADED/REFRAMED through v0.6: - Wolfenstein hierarchy substrate source: from “su(3) composite root forward” (K401, K402) → “falsifiable check seats must pass; source gated on T_3^R mechanism” (K403, K404) - Cabibbo via Gatto-Sartori-Tonin: from “forward cross-check 0.62% tight” (K400, K401) → “GST relation with observed mass input; falsifiable target seats must hit; not banked fingerprint” (K402) - λ_3 identification: from “convergence” framing (K400, K401) → “recognition of standard su(3)” (K402) - “Residual 0.413 = $|\rho - i\eta|$ ” CP magnitude framing: from “what should be there per Jarlskog” (K401, K402, K403) → “parametrization algebra not derivation” (K404) - M-question scorecard: from “M=1 → R=3 still strong” (K401, K403) → “M splits; even $M_{\text{angle}}=0$ caps at 3 angles; M=0 strict bar; no escape hatch” (K404)

6. Atlas tier state at v0.6 close

Observable	Tier
$\alpha = 1/137$	FORCED DEEP (founding)
$\theta_{\text{QCD}} = 0$	FORCED DEEP (founding)
Muon $(24/\pi^2)^6$	FORCED DEEP (banked clean K388)
Tau leading $49.71 + \sqrt{\pi}$ open	FORCED YELLOWER PROVISIONAL (un-bank condition K390)
α^{12} hierarchy coefficient = C_2	Tier-2 IDENTIFIED (K393 said-and-stands)
Lightest ν mass m_1	0 mathematical (K396 F159 Casey-sharpened)
ν locus + Γ_{Ω} pole	Forward characterized (K399 F158)
Locus-difference principle	Forward structural-grade (K403; verified K404 by Grace 2.6× to 43×)
Banked BST over-determination fingerprints for #418	ZERO (K404 Grace synthesis)

Count HONESTLY 4 of 26 strict (3 deep + 1 provisional with un-bank condition tracked). Ledger v1.81.

7. v0.6 close

v0.6 SUPPLEMENT absorbs K398-K404 (7 K-audits Tuesday EOD + Wednesday morning + #418 checkpoint cascade): - 6 Keeper self-corrections in cascade (13th, 14th, 15th, 16th, 17th, 18th) - 5 new Keeper discipline forcing functions (FF-14 through FF-18) - Substantial Wednesday morning thermodynamic cycle closure (F165-F176 absorbed via K399) - Substantial Wednesday late-morning #418 quark-sector checkpoint cascade (K400-K404) - Locus-difference principle banked at structural-grade - Banked over-determination fingerprints for #418 today: ZERO

Total K-audit chain at v0.6 close: 234 K-audits.

— Keeper, Wednesday 2026-06-17 13:00 EDT (date-verified) — K-Audit Registry v0.6 SUPPLEMENT: K398-K404 indexed; 234 K-audits total; 5 new forcing functions FF-14 through FF-18; count HONESTLY 4 of 26; locus-difference banked at structural-grade